



Tool for R&I of the 797F High Pressure Fuel Pump

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Mining equipment are part of the production process, with an important role in meeting the production targets of any mining operation. Maintenance and repair tasks must be executed in order to ensure the availability and reliability of the equipment, and these must be executed efficiently, with quality and safety.

In large mining equipment, such as the Caterpillar 797F trucks, these tasks include handling of large and heavy components.

In order to meet the safety, efficiency and quality objectives, the maintenance department must ensure that its staff has the necessary training, support information (technical and procedures), infrastructure, tools and support equipment.

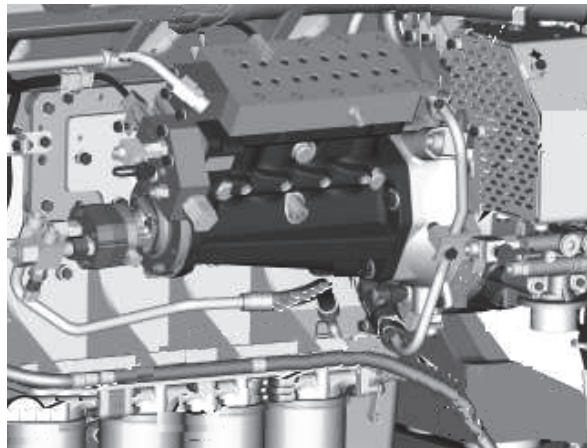
This Best Practice refers to the design, fabrication, use and benefits of a tool developed to remove and install the high-pressure fuel pump in 797F trucks, in order to contribute towards meeting the objectives mentioned above.

2.0 Description of the Best Practice

2.1 Detection of the Opportunity

During the analysis of the remove & install of the high-pressure fuel pump in the 797F trucks, 4 areas of opportunity for improvement were identified. These opportunities motivated the development of the accessory proposed in this Best Practice.

Areas of Opportunity



Safety

The traditional procedure recommended for the Removal & Install of the steering accumulators (Media # KSNR6052-11), includes the use of slings/straps, lever puller tool and adequate hoist equipment. Due to the sensitive nature of the procedure, successful application relies on the experience of the technician (s) performing the task. The technician (s) must maintain control and balance of the Fuel Pump at all times to avoid the risk of damage to the part or equipment from striking the part during movement.

Any work that includes a “suspended” load is risky by nature. Special attention must be given to safety, specifically avoiding pinch points and contacting or striking the technician(s) with the swinging Fuel Pump. The High Pressure Fuel Pump weight: 157 kg.

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Time (Efficiency)

In accordance with our records / repair history, the time required for this task (R&I of the Fuel Pump), following the former recommended procedure, is on average 14 hours of machine down time.

Resources (Labor)

Normally, 3 technicians are assigned to execute this task. This means that we use 42 man hours in total to complete the task following the former recommended procedure.

Quality

Some events of sub-standard quality have been observed and recorded during the R&I of the Fuel Pump. These events have generally been attributed to the lack of control of the Fuel Pump while it is being manipulated as a suspended load

2.2 Proposed Improvement Solution

The proposed solution is the use of a lifting tool, specially designed for this application (Image 1), the HRIBC-DMH (Initials in Spanish that stands for: Tool for R&I the Fuel Pump – Division Ministro Hales).

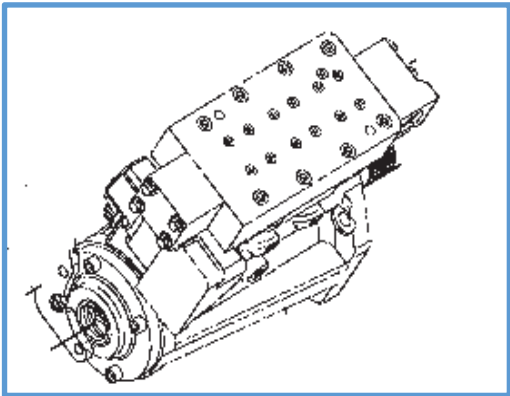


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High-pressure fuel pump

This Tool provides us with the safety and maneuverability necessary to carry out the task with significantly improved control. This solution changes the original situation of working with a suspended load to one with a supported load, which is much more favorable regarding the safety of the maneuver.

The following photograph shows the HRIBC-DMH Tool operating in our workshops.



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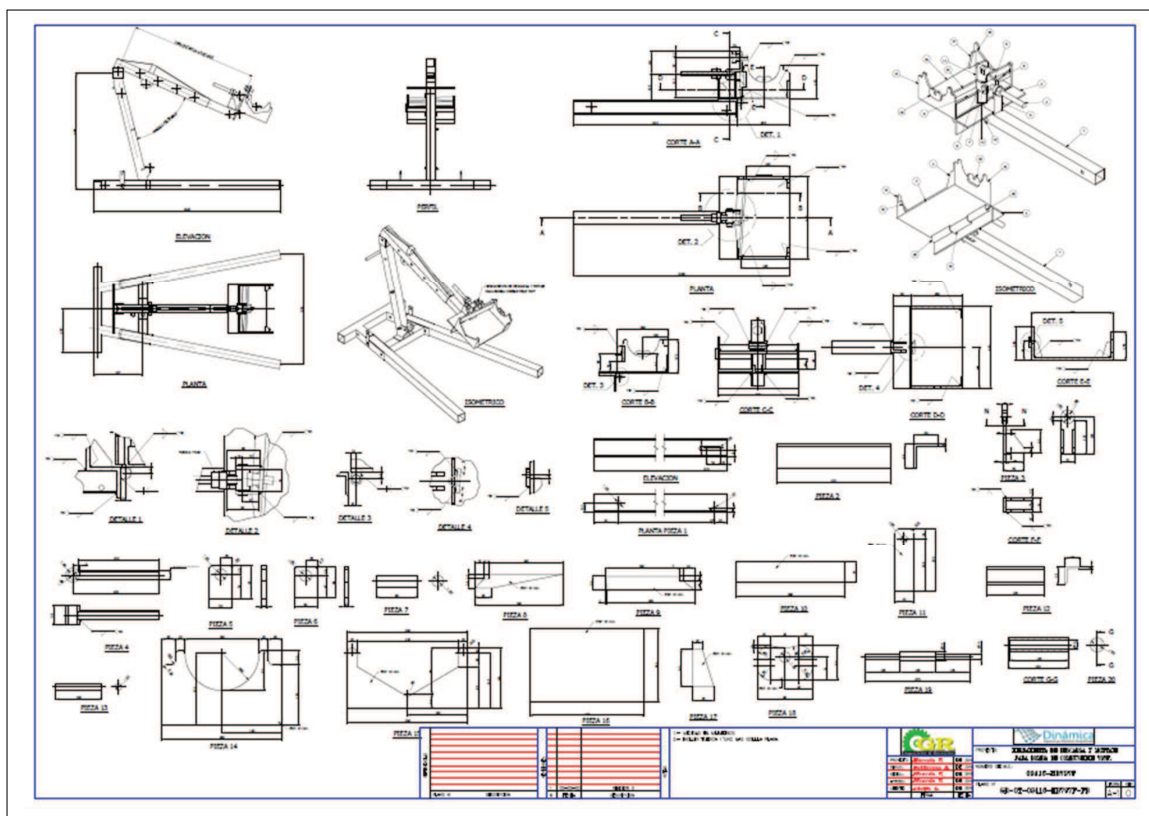
3.0 Steps for implementation

The general steps to follow for the implementation of this best practice are:

- 3.1 Fabrication of the HRIBC-DMH
- 3.2 Technical certification of the HRIBC-DMH
- 3.3 Practical tests in R&I of the High Pressure Pump
- 3.4 Final adjustments
- 3.5 SOP (Standard Operating Procedure) Documentation
- 3.6 Personnel Training
- 3.7 Start-up and Validation of Benefits

3.1 Fabrication of the HRIBC-DMH

The ARIA-DMH was fabricated locally, following the design specifications indicated below. This design was carried out locally in the Finning – DMH contract.



This fabrication drawing is attached to this Best Practice.

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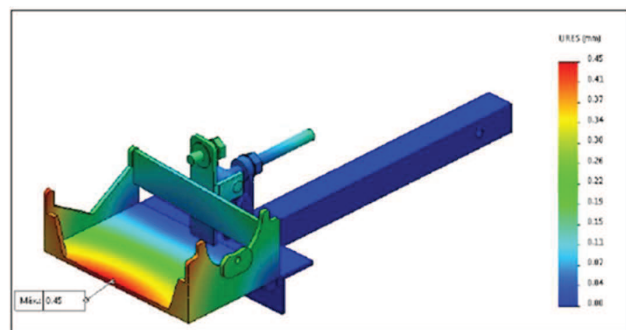
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3.2 Technical Certification of the Tool for R&I Fuel Pump (Welds and Capacity)

The certification of the welds and analysis of the capacity of the accessory once fabricated is important.



Structural Calculations Report
and Quality Certification of the
HRIBC-DMH



Example of tests / stress analysis conducted on
the HRIBC-DMH

3.3 Practical Tests on Tasks involved in R&I of the Fuel Pump

Once the Tool has been received, we recommend conducting operating tests at the workshop in a controlled and safe environment, with the objective of identifying and carrying out the final adjustments if required.

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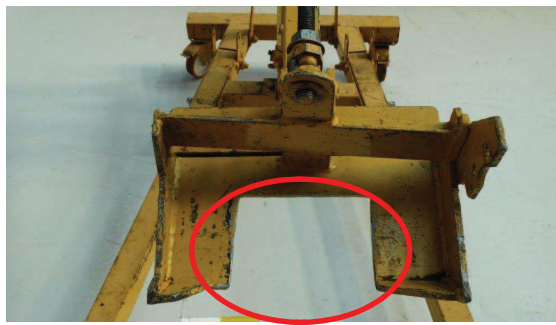
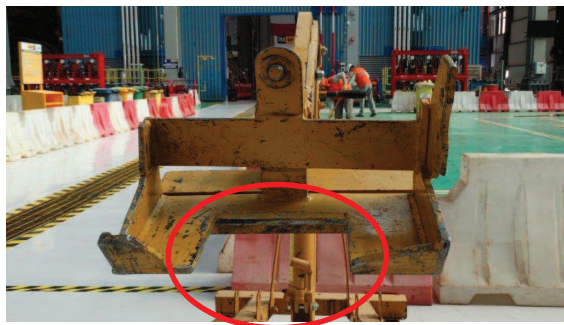
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3.4 Final adjustments

Adjust what is detected and required.

In our case it was detected that when the pump was installed on the support tool's base, it could be destabilized. To resolve this problem we made a cut in the tool base, with which the pump was better supported, as shown in the following photographs.

Modification of the support base



3.5 Documentation of the SOP (Standard Operating Procedure)

In order to ensure the proper operation of this tool, we recommend documenting the standard operating procedure (SOP).

We also recommend reviewing the local safety regulations that applies to your operation in order to adapt the SOP to its specific needs.

3.6 Employee Training

Before releasing the HRIBC-DMH to normal operation it is important to train the technicians that are going to use it regularly. Consider training at least 2 technicians per shift.

It is recommended to include the application of the SOP in the training, as well as hands on practice using the HRIBC-DMH in a controlled and safe environment.

3.7 Start-up and Validation of Benefits

Once the HRIBC-DMH gets released to operation we recommend following up closely with Fuel Pump R&I events to ensure the correct use of the tool as well as the documentation of the benefits.

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Once the benefits have been established for your operation, we recommend updating Standard Jobs maintained and applied by Planning & Scheduling department with the new time and labor (man hour) estimates for this job.

4.0 Benefits

In accordance with our observations and records regarding the use of this accessory in our operation, the benefits documented to date reflect the following improvements.

	Risk Level without HRIBC-DMH	Risk level with HRIBC-DMH	Number of Technicians without HRIBC-DMH	Number of Technicians with HRIBC-DMH	Total machine time without HRIBC-DMH	Total machine time with HRIBC-DMH	Man hours without HRIBC-DMH	Man hours with HRIBC-DMH
Change of Fuel Pump	High	Controlled - Low	3	2	14	7	42	14

As seen in the table of results, benefits are found in the areas of Safety, Task Execution Time, Use of Labor and Quality of Execution.

Safety: With the use of the HRIBC-DMH you will change the work mode from “suspended load” to “supported load”, and will also add a level of control to the whole task. These two factors will reduce the level of risk for this task.

Execution Time The time reduction we recorded is up to 50 %, meaning that the total DT of the machine will be 7 hours vs. 14 hours former required for this task.

It is important to mention that the reduction of down time will directly influence fleet MTTR results. The final influence on the MTTR results will have to be determined based on the frequency of the accumulator change.

Man Hours The reduction of 1 support person means a total reduction of Labor Hours of 28 hours, a 67 % reduction, which can be allocated to a different task. This labor reduction will ultimately influence positively the results of MR (maintenance ratio) of the equipment fleet.

Quality: Even though it is difficult to quantify, it is evident that by improving safety, the level of control of the task and the technician effort, these factors will positively influence the quality of the job done. Any quality improvement to tasks will positively influence MTBS of the equipment.

5.0 Resources Required

The resources required are those specified in this Best Practice.

6.0 Additional Support / References

The following information is attached to this Best Practice

- Fabrication Diagram / Blueprints

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7.0 Other Associated Best Practices

N/A

8.0 Acknowledgements

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